MEDIANTS AND BOOSTS ON ONE PAGE

Merging and boosting

Write each speed as momentum over energy: the mediant merges two bodies and halves rapidities, Einstein's rule changes frame and adds them.

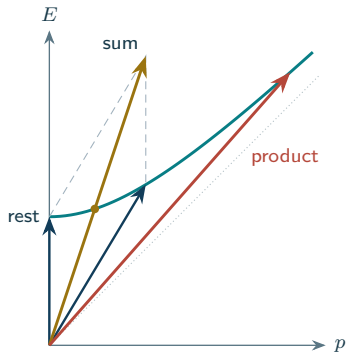
THE RULE

A speed is a fraction $\beta = p/E$

With $c = 1$, a body of mass m and rapidity φ has energy $E = m\cosh\varphi$ and momentum $p = m\sinh\varphi$, so its speed is the fraction $\beta = \frac{p}{E} = \tanh\varphi$ and its energy–momentum is a vector on the hyperbola $E^2 - p^2 = m^2$. Two such vectors combine in two ways:

$$\frac{p_1}{E_1} \oplus_{\mathrm{m}} \frac{p_2}{E_2} = \frac{p_1 + p_2}{E_1 + E_2}, \qquad \beta_1 \oplus_{\mathrm{E}} \beta_2 = \frac{\beta_1 + \beta_2}{1 + \beta_1\beta_2}$$

The mediant adds the vectors and gives the velocity of the merged pair, its centre of momentum, which always lies between the two; Einstein's rule composes boosts, changing frame, and adds the rapidities.



Navy: at rest and at $\varphi = \ln 2$. Their sum points along $\frac{1}{2}\ln 2$; the product of the moving one with itself reaches $\ln 4$.

HALF AND DOUBLE

Equal masses: the mediant averages rapidities exactly

For equal masses the sum-to-product identities give

$$\frac{\sinh\varphi_1 + \sinh\varphi_2}{\cosh\varphi_1 + \cosh\varphi_2} = \tanh\frac{\varphi_1 + \varphi_2}{2},$$

so the merged pair moves at the average rapidity, while a boost moves to the sum. Merging with an identical body at rest therefore gives the half-rapidity formula and boosting twice gives the double-rapidity formula:

$$\frac{0}{1} \oplus_{\mathrm{m}} \frac{\gamma\beta}{\gamma} = \frac{\gamma\beta}{1+\gamma} = \tanh\frac{\varphi}{2}, \qquad \beta \oplus_{\mathrm{E}} \beta = \frac{2\beta}{1+\beta^2} = \tanh 2\varphi$$

starting from $\beta = \frac{3}{5}$, $\varphi = \ln 2$	merge with rest	merge with a twin	boost by itself	boost by its reverse
operation	$\frac{0}{1} \oplus_{\mathrm{m}} \frac{3/4}{5/4}$	$\frac{3/4}{5/4} \oplus_{\mathrm{m}} \frac{3/4}{5/4}$	$\frac{3}{5} \oplus_{\mathrm{E}} \frac{3}{5}$	$\frac{3}{5} \oplus_{\mathrm{E}} \left(-\frac{3}{5}\right)$
rapidity	$\frac{1}{2}\ln 2$	$\ln 2$	$\ln 4$	0
speed	$\frac{1}{3}$	$\frac{3}{5}$	$\frac{15}{17}$	0
mass afterwards	$\sqrt{4.5}\,m = 2.12m$	$2m$	m	m

UNEQUAL MASSES AND THE ENERGY RELEASED

Denominators are weights, and merging costs nothing but makes mass

For unequal masses the mediant is weighted by energy, so writing $\frac{3}{5}$ as $\frac{0.75}{1.25}$ or $\frac{7.5}{12.5}$ matters: the denominator says how heavy the body is. The merged mass comes from the Minkowski length of the summed vector,

$$M^2 = (E_1 + E_2)^2 - (p_1 + p_2)^2 = m_1^2 + m_2^2 + 2m_1m_2\cosh(\varphi_1 - \varphi_2) \;\geq\; (m_1 + m_2)^2,$$

the reverse triangle inequality: the faster the two approach, the more kinetic energy becomes rest mass. Two protons meeting head-on at $\pm\frac{3}{5}$ have relative rapidity $\ln 4$ and give $M = 2.5\,m_p$; a beam hitting a proton at rest needs the same relative rapidity, the Einstein double $\frac{3}{5} \oplus_{\mathrm{E}} \frac{3}{5} = \frac{15}{17}$, to reach the same M, which is why colliders beat fixed targets.

THE CIRCULAR TWIN

Adding unit vectors bisects the angle

On the circle the same move gives the half-angle formula, $\frac{0}{1} \oplus_{\mathrm{m}} \frac{\sin\theta}{\cos\theta} = \frac{\sin\theta}{1+\cos\theta} = \tan\frac{\theta}{2}$, because two unit vectors span a rhombus whose diagonal bisects their angle; multiplying them as complex numbers doubles the angle instead. In both geometries the mediant of unit vectors halves and the product doubles.

COMPOSING THE TWO MOVES

Boosts commute with merging

Boosting every body by ψ adds ψ to every rapidity and to the merged one, so merge-then-boost equals boost-then-merge: $(\beta_1 \oplus_{\mathrm{m}} \beta_2) \oplus_{\mathrm{E}} \beta_\psi = (\beta_1 \oplus_{\mathrm{E}} \beta_\psi) \oplus_{\mathrm{m}} (\beta_2 \oplus_{\mathrm{E}} \beta_\psi)$ with each fraction carried as p/E. Galileo's limit keeps both moves, with mass-weighted mean velocity for merging and plain addition for boosts.

Conventions. Units with $c = 1$; $\oplus_{\mathrm{m}}$ is the mediant of fractions written as momentum over energy and $\oplus_{\mathrm{E}}$ is Einstein's rule; $\gamma = \cosh\varphi$ and $\gamma\beta = \sinh\varphi$.

Constructed by Silicon · Anthropic Claude — Architect Carbon · LeeMcCullochJames